

Integrating generative artificial intelligence into chemical engineering education: Staff and student perspectives

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ARTICLE INFO

Keywords:

Large language models
Digital tools
Adaptability

ABSTRACT

Generative artificial intelligence (GAI) is rapidly transforming higher education, raising pedagogical, ethical, and epistemological challenges. In chemical engineering, concerns have emerged that student over-reliance on GAI may undermine the development of key skills such as problem-solving, creativity, and critical thinking. This mixed-methods study explored staff and student perspectives on the integration of GAI into chemical engineering education. Data were gathered through questionnaires (students $n = 115$; staff $n = 17$), two student focus groups, and five semi-structured staff interviews. Quantitative data were statistically analysed, and qualitative data were examined using reflexive thematic analysis. Five themes were identified: ethics, reliability and accuracy, impact on learning, pedagogical disruption, and staff use. Applying the Technological Pedagogical Content Knowledge (TPACK) model highlighted distinct staff–student differences. Students viewed GAI as a beneficial learning support, with limited concern for bias or authorship. Staff raised critical issues around reliability, transparency, and pedagogical alignment, but acknowledged GAI's potential to streamline routine tasks such as feedback provision. Based on these insights, this study goes beyond reporting perspectives, to also propose curriculum interventions including early AI literacy and critique-based assessment. It also offers policy recommendations addressing equity, sustainability, and the responsible integration of GAI into chemical engineering education.

1. Introduction

Since the release of accessible large language models (LLMs), largely stemming from the launch of ChatGPT v3.5 in November 2022, generative artificial intelligence (GAI) has become one of the most significant disruptors to higher education (HE) in modern times (Kanbach et al., 2024). Its widespread availability raises concerns over authorship, trust, and equity which might negate the potential opportunities this technology represents with regards to possible improvement in coursework feedback, student support, and administrative tasks (Dotan et al., 2024). Although high level information may be accessed quickly, there are currently some limitations of GAI, especially when applied to subject-specific areas. As an example, LLMs often lack the ability to complete complex calculations, cannot perform logical reasoning tasks, and are also unable to carry out the practical aspects of process design or simulation (Keith et al., 2025). As GAI develops, it is possible that these limitations will be overcome. Further, as AI tools become more seamlessly integrated into background computer operating systems, it will only become further necessary to position GAI as a learning support

structure, rather than it replacing the development of specific skills (Moorhouse et al., 2023). For example, passive use of an AI tool to consolidate complex information from multiple sources, risks people, particularly students, missing the opportunity to acquire a deeper understanding of subject-specific knowledge by conducting independent research and summarising relevant information (Tan & Maravilla, 2024). To address these challenges whilst ensuring that graduates possess the skillsets needed to thrive in an AI-enabled workplace, Universities are being forced to re-evaluate teaching and learning practices, and revise academic integrity policies. Whilst numerous studies have been completed looking at staff and student perceptions of GAI in higher education (Chan & Zhou, 2023; JISC, 2023; Kim et al., 2025), there is at present a lack of course-specific evidence to guide the strategy of individual disciplines (Wang et al., 2024). The work presented aims to address this gap by taking a mixed-methods approach to identify the perceptions of staff and students specifically within the School of Chemical Engineering. Through this lens, the research findings may be used to inform a strategy for the integration of GAI education into existing chemical engineering curricula across different institutions. To

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<https://doi.org/10.1016/j.ece.2026.100503>

Received 5 November 2025; Received in revised form 19 December 2025; Accepted 29 January 2026

Available online 31 January 2026

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provide the context for this research, the remainder of the introduction considers the wider adoption of GAI into higher education (including institutional governance and policy alignment), ethical implications, and the role of educational frameworks before focussing on the literature available which specifically considers the use of GAI in chemical engineering education.

1.1. Adoption of GAI in higher education

The integration of GAI, especially LLMs, into HE is being driven by the necessity of AI literacy within subject-specific contexts (Chan and Colloton, 2024). Research has explored the ethical risks, changes to pedagogy, and potential suitable applications (Batista et al., 2024; Dotan et al., 2024; Teräs, 2022). Later work highlighted the importance of AI literacy and intelligent Technological Pedagogical Content Knowledge, identifying how perceived trust is a key influence on educators' willingness to adopt GAI (Al-Abdullatif, 2024). Similarly, Celik (2022) emphasised how successful integration relies on an educators' awareness and their ability to adapt (Celik et al., 2022). These studies suggest that GAI's educational value depends on institutional support, structured training for staff, and digital readiness (Al-Abdullatif, 2024; Celik et al., 2022; Wang et al., 2024). Fortunately, as GAI use becomes more prevalent, many institutions have published their own guidance and policies on how GAI may be used to support teaching and learning. In turn, these policies and different approaches have been subjected to some academic scrutiny. Wang et al. (2024) analysed strategic responses from global universities, identifying multi-tiered, role-specific responses to manage GAI integration (Wang et al., 2024). Belkina et al. (2025) found that academic integrity, equitable access, and cross-disciplinary coordination are global trends within institutional policies (Belkina et al., 2025). However, the current policy frameworks are generally high-level, with minimal course-specific guidance (McDonald et al., 2024). The unresolved issues for STEM disciplines include the traceability of AI-assisted calculations and specific training needs. Until institution-level policies become more tailored to each discipline, there is a risk that both pedagogical innovation and ethical oversight may be undermined (Chen et al., 2023; Mendez, 2024).

1.2. Ethical and epistemological implications

When developing appropriate strategies to support GAI integration into various curricula, it is of paramount importance to consider both the ethical and epistemological implications of its use. As GAI becomes increasingly prevalent, ethical considerations must become prominent to effective implementation. Dotan et al. (2024) argued that alignment with academic values and institutional integrity are key "points to consider" for responsibly introducing GAI (Dotan et al., 2024). Tan and Maravilla (2024) suggested that, if GAI is embedded thoughtfully, it can improve academic honesty: promoting ethical awareness between students (Tan and Maravilla, 2024). However, these interpretations often generalise disciplines, broadly covering ethical concerns compared to nuanced issues within STEM where ethical risks go beyond just plagiarism (Chen et al., 2023; Mendez, 2024). One of the more concerning cases reported involved AI hallucinations. In a trial, the LLM ChatGPT v3.5 completely fabricated information within a Reporting of Injuries, Diseases, and Dangerous Occurrences Regulations (RIDDOR) report, meaning that an incident's causes, remedial actions, and learning opportunities were all false (McDermid, 2023). These hallucinations have previously been highlighted as a major ethical risk due to the potential for serious safety implications (Ji et al., 2023). In turn, there are likely to be legal consequences should an individual or business be found to have used LLMs (or GAI more broadly) in a similarly irresponsible manner. To address some of these concerns, it is possible to apply existing educational frameworks to the relatively new technology of LLMs, as described in the following section.

1.3. Educational frameworks

Educational frameworks are structured sets of principles, guidelines, and practices which guide the design, development, and delivery of teaching and learning activities. They often contain a vision to identify the overarching educational goal, specify a curriculum, define methods for effective instruction and student engagement, and outline how learning will be assessed (Conole, 2019). Constructivism, particularly Vygotsky's Zone of Proximal Development (ZPD), emphasises that learners actively construct knowledge through interaction and reflection. GAI tools, therefore, can act as scaffolds to support student learning (Schunk, 2012; Vygotsky, 1980). Also of relevance here is the Technological Pedagogical Content Knowledge (TPACK) framework developed by Mishra and Koehler (2006). This defined that effective technology integration is at the intersection of content expertise, pedagogical strategies, and technological knowledge (Mishra and Koehler, 2006). For STEM disciplines such as chemical engineering, the need for content accuracy and reproducibility increases the relevance of applying this model. Both constructivism and TPACK hence provide a lens through which to analyse how staff perceive GAI's potential in topics such as content delivery, feedback design, and curriculum development (Al-Abdullatif, 2024; Keith et al., 2025) as discussed in later sections of this work.

1.4. GAI in chemical engineering education

As technologies advance, it is essential to ensure that graduates are equipped with the ability to continuously develop their own skillsets and hence become lifelong learners. A core competency of chemical engineering graduates in particular is critical thinking, a skill which will only become more important as GAI evolves. It is therefore necessary to ensure that chemical engineering students can independently evaluate the output of an LLM. As such, it is therefore desirable to design alternative assessments which explicitly ask students to use an LLM (Ravi, 2023). It has also been previously observed that, by carefully integrating GAI technologies into chemical engineering curricula, students may be able to learn beyond the initial problem set and thus gain a deeper, more holistic understanding of a given concept, whilst becoming more critical of their own work. However, this same research does also warn that LLM outputs in particular need to be carefully checked for accuracy (Tsai et al., 2023).

There have been some specific case studies reported where LLMs have been successfully integrated into chemical engineering curricula. One such example by Keith et al., utilised the IDEE (Identify learning outcomes, Determine automation level, Ensure ethics, and Evaluate effectiveness) framework to build the use of ChatGPT v3.5 into existing lab sessions. This enabled students to build AI literacy by creating and evaluating the output of the LLM, and hence illustrates how GAI can be used to support learning in technical scenarios if interacted with critically (Keith et al., 2025). This critique is essential to ensure that outputs are thoroughly evaluated, and not simply taken at face value (Mendez, 2024). Without this structured support for analytical thinking or ethical reflection, students may risk not developing core principles. Although it is clear from works such as these that GAI is being integrated into chemical engineering education, there is a lack of discipline-specific research regarding the perceptions of both staff and students. Recent publications tend to identify student experiences or generalise GAI's impact across HE without analysing specific disciplines (Batista et al., 2024; Belkina et al., 2025; Chan and Colloton, 2024; Essel et al., 2024). Critically, few studies examine both staff and students' perspectives simultaneously, especially in fields such as chemical engineering (Guillén-Yparrea and Hernández-Rodríguez, 2024; Kim et al., 2025). When ethical, pedagogical, and institutional dimensions are discussed, limited framing of educational theories is introduced (Al-Abdullatif, 2024; Williamson and Eynon, 2020). This work aims to address this gap by answering the specific research questions detailed in the following

section.

1.5. Research questions

This study aims to explore staff and student perceptions of GAI in chemical engineering education and therefore is guided by the following research questions:

1. How do staff and students perceive the opportunities and risks of using GAI in chemical engineering education?
2. What similarities and differences emerge between staff and student perceptions?
3. How can Constructivism and TPACK enhance our understanding of GAI's educational impact?
4. What institutional strategies can help support the responsible integration of GAI?

In order to answer these questions, this study takes a mixed-methods approach as described in the following section.

2. Methodology

2.1. Pedagogical approach

A mixed-methods design was applied, integrating both quantitative and qualitative approaches aligned with the Constructivism and TPACK frameworks, which became central lenses within the analysis (Mishra and Koehler, 2006; Piaget, 1952). TPACK was used to interpret staff perceptions regarding content expertise, pedagogical strategies, and technological proficiencies with GAI tools (Mishra and Koehler, 2006). This design was chosen to provide a balanced and rigorous understanding of both staff and students' perceptions, enabling comparison and validation across different data sources (Creswell and Plano Clark, 2007). It enhanced the study's credibility by combining statistical and qualitative insights. The integration of both data types linked emerging qualitative themes with statistically significant trends, ensuring the reliability of recommendations (Selwyn, 2019).

2.2. Data collection

Two separate questionnaires were designed for staff and students within the School of Chemical Engineering at the University of Birmingham. Both were informed by recent literature in HE and aligned with Constructivist and TPACK frameworks (Mishra and Koehler, 2006; Piaget, 1952). These questionnaires are available in full in the [Supplementary materials](#) but in brief, they covered assessment of participants' familiarity with GAI, and perceptions of its educational value, applications within chemical engineering, and potential challenges and concerns (Batista et al., 2024; Creswell and Plano Clark, 2007). Questionnaires were constructed in Microsoft Forms and distributed via the University's virtual learning environment, teaching faculty email lists, and direct in-person promotions to improve participation (Selwyn, 2019). Pilot tests ensured the clarity and relevance of questions, and additional incentives were created for students with the opportunity to win a £ 50 gift card, to maximise participation (Creswell and Plano Clark, 2007; Schunk, 2012).

Additional qualitative data were collected through focus groups and interviews. Two focus groups (Year 1–2 and Year 3–4) of five participants per group were conducted to gain students' insights, along with five one-to-one semi-structured interviews with academic staff. Focus groups captured peer-based dialogue, while interviews enabled deeper, individual reflections from staff. The purely qualitative data collection methods all ran for 30–60 min, were transcribed verbatim, and anonymised.

Before participation, informed consent was acquired through the use of participant information sheets and consent forms, where the

objectives, procedures, confidentiality measures, and participants' rights were stated. Participation was voluntary, anonymous, and confidential to adhere to the University's ethics policy.

2.3. Quantitative data analysis

Quantitative data were processed in Microsoft Excel and statistically analysed in Minitab. A reliability analysis (Cronbach's α) ensured the internal consistency for grouped Likert items, supporting the composite analysis (Field, 2013). A linear regression model tested whether perceived educational benefits correlated with the intended future use of GAI, compared to quantified desire for future use. Residuals were calculated as the difference between students' actual likelihood rating and the predicted value, which formed the basis for the model analysis. A normal probability plot checked the residuals' normality, with a residuals versus fitted plot being obtained to understand the homoscedasticity (constant variance) (Field, 2013).

Statistical significance between staff and student was determined through independent *t*-tests and Cohen's *d* (effect size) calculations were conducted to assess the practical significance of the data. All statistical tests were conducted at a significant threshold of $p < 0.05$ (Field, 2013). Final visualisations were generated using MATLAB to improve the clarity of the data.

2.4. Qualitative data analysis

Qualitative data were analysed applying Braun and Clarke's (2006) reflexive thematic analysis, using NVivo 14 (Braun et al., 2022). NVivo supported systematic coding and theme development across all qualitative data. Initial codes were collaboratively refined with a second researcher to establish an agreed coding framework, enhancing inter-rater reliability (Campbell et al., 2013).

This methodological design enabled the collection of credible, actionable insights to inform targeted recommendations for GAI integration into the chemical engineering curriculum. Vygotsky's Constructivism provides a strong lens through which to understand student engagement with GAI, being an active, scaffolded process to knowledge construction (Piaget, 1952; Vygotsky, 1980). Similarly, TPACK helped frame the complexities of staff responses around teaching using technology, subject expertise, and pedagogical goals (Mishra and Koehler, 2006). This theoretical awareness enhanced the study's reflexivity and quality of interpretations from findings (Braun et al., 2022).

3. Results and discussion

Following the mixed methods approach outlined in [Section 2](#), a reflexive thematic analysis of staff and students' responses to the surveys, semi-structured interviews, and focus groups revealed six distinct themes: Ethical concerns, reliability and accuracy of AI outputs, impact on student learning, pedagogical and assessment practices, staff use and application of AI, and AI being the next shift in technology. These were informed by the agreed coding framework developed by two of the researchers. This framework, along with example quotes which informed them, is presented in [Table 1](#). Throughout the remainder of this paper, each of these themes are discussed in detail with the quantitative survey data used to provide additional, statistically significant insight into the findings obtained. Additional survey data regarding respondent demographics is available in the [Supplementary Materials](#).

3.1. Theme 1: ethical concerns

Ethical concerns surrounding GAI within HE emerged as a multifaceted theme for both staff and students. Four key issues were identified: algorithmic bias, copyright and IP, environmental impact, and equity of access. The concerns identified align with broader discourses

Table 1
Coding framework of participant responses, grouped according to theme.

Theme	Code	Quote	Source
1. Ethical Concerns	Bias in AI output	“Definitely a propensity for bias...if it’s been trained on pre-existing work”	Staff 1
		“You need the skillset to interrogate the output...and understand its inbuilt bias”	Staff 2
	Copyright & IP	“Why are you allowed to prompt this software, with no acknowledgement of royalty?”	Staff 3
		“Still using unholy amounts of energy every time we ask it a question”	Staff 1
	Environmental & resource impact	“My big concern is the energy and water use of AI”	Staff 3
		“Massively unfair that a rich kid can get a better AI model than a poor kid”	Staff 1
	Equity & accessibility	“The University has a duty to make AI accessible”	Staff 2
		“AI coming up with answers that are illegal or irresponsible...no ethical filter”	Staff 4
	Lack of ethical responsibility	“GAI models aren’t very good at consolidating lots of information”	Staff 3
		“You don’t necessarily know that the information it spits back at you...it still might hallucinate the citation”	Staff 5
2. Reliability & accuracy of AI outputs	Limited functionality	“Doing it by yourself and critiquing what you get out – probably a valuable skill going forward”	Y3 & Y4 focus group
		“I wouldn’t want something that involves error to mark my work presently”	Y1 & Y2 focus group
	Lack of transparency	“GAI can generate something which looks good but is in fact nonsense”	Staff group
		“Working smartly with AI rather than letting it do the work...probably the best way”	Staff survey
	Need for critical evaluation of output	“If you don’t understand thermodynamics, you might allow something to go forward which just can’t work”	Y3 & Y4 focus group
		“Worried that Y1 and 2 students might just copy from GAI without learning the skills”.	Staff 2
	Lack of trust	“Built up a debt in first and second year due to AI reliance”	Staff 5
		“Always criticise the output, they will be designing dangerous things otherwise”	Staff 2
	Need for validation of AI output	“Sometimes provides code for a more advanced version... like Matlab”	Staff 5
		“Students are always going to use it”	Student survey
3. Impact on student learning	AI as a tool for students	“Must be instruction on how to use it effectively and how to recognise and correct bias”	Y1 & Y2 focus group
		“Ask GAI how to answer a coursework question, then	Staff 4
	Need for foundational knowledge		Staff 4
			Staff 5
	Risk of de-skilling		Staff 2
			Staff 5
	Skill development		Staff 5
			Student survey
	Student expectations		Y1 & Y2 focus group
			Staff survey
4. Pedagogical and	Authentic project based work		Staff 4
			Staff 4

Table 1 (continued)

Theme	Code	Quote	Source
assessment challenges	critique it – which I think is a nice approach”	“You can see aspects of assessment being pushed back to more exam style”	Staff 2
		“Over-reliance on AI from both students and staff is the main concern”	Staff 3
	Over-reliance on AI	“If we could develop an AI tool trained on all of our teaching material, that could be really valuable”	Staff 3
		“It invites cheating left and right”	Staff survey
	Personalised teaching assistant	“Requires clear policies and guidelines, such as proper citation practices”	Staff survey
		“GAI can give all the feedback on the formatting and structure – I can focus on the technical”	Staff 5
	Plagiarism & academic integrity	“It should make certain aspects much quicker, not having to read thousands of papers”	Staff 2
		“Staff members need to be trained first”	Staff survey
	Policies & guidelines for AI	“An AI tool will never replace the benefits of human-to-human interaction”	Staff 3
		“Even if it takes me longer, I want to know that I set that up how I intended”	Staff 5
5. Staff use and application of AI	Assessment support	“Misused GAI without proper oversight will produce sub-par educational content”	Staff survey
		“Proper training...on responsible AI usage will be essential”	Staff survey
	Impact on staff workload	“I see the release of ChatGPT in a similar light to people who... might have seen the release of Google”	Y3 & Y4 focus group
	Lack of staff knowledge		
	Need for human interaction		
	Staff resistance to AI		
6. Next shift in technology	Generating teaching material		

on ethical AI use, such as emphasising fairness, responsibility, and sustainability (Farazouli et al., 2024; Tan & Maravilla, 2024; Zawacki-Richter et al., 2019).

A disparity between staff and students’ perspectives was highlighted from the survey results. Overall, 45.2 % of students expressed concern (Fig. 1a), compared to 100 % of staff (Fig. 1b). Understanding this is further explored in the analysis of the qualitative data below. Masters’ students (Year 4 or 5) disrupted the decreasing trend in concern (Fig. 1c), which may reflect the minimal exposure or dependency on generated outputs since GAI’s release in 2022, or the different quality of work which is required by them. However, it is worth noting that only 10 respondents were at this level, with the majority completing their first two years of the course. This means that a small number of students expressing concern for GAI will have a large influence on the proportion of the total. Reassuringly, over 50 % of students did recognise that GAI outputs may be prone to bias (Fig. 1d), with 12 % remaining neutral, indicating some developed awareness that the output may need to be critiqued and / or cross-checked against other, more traditional sources of information (Baidoo-Anu and Owusu Ansah, 2023; Hammond et al., 2024).

This concern regarding the possible bias in GAI outputs was further reflected in the qualitative data obtained, with this issue repeatedly raised by both students and staff. It was not just viewed as a technical issue, but a deeper epistemic risk jeopardising fairness and knowledge

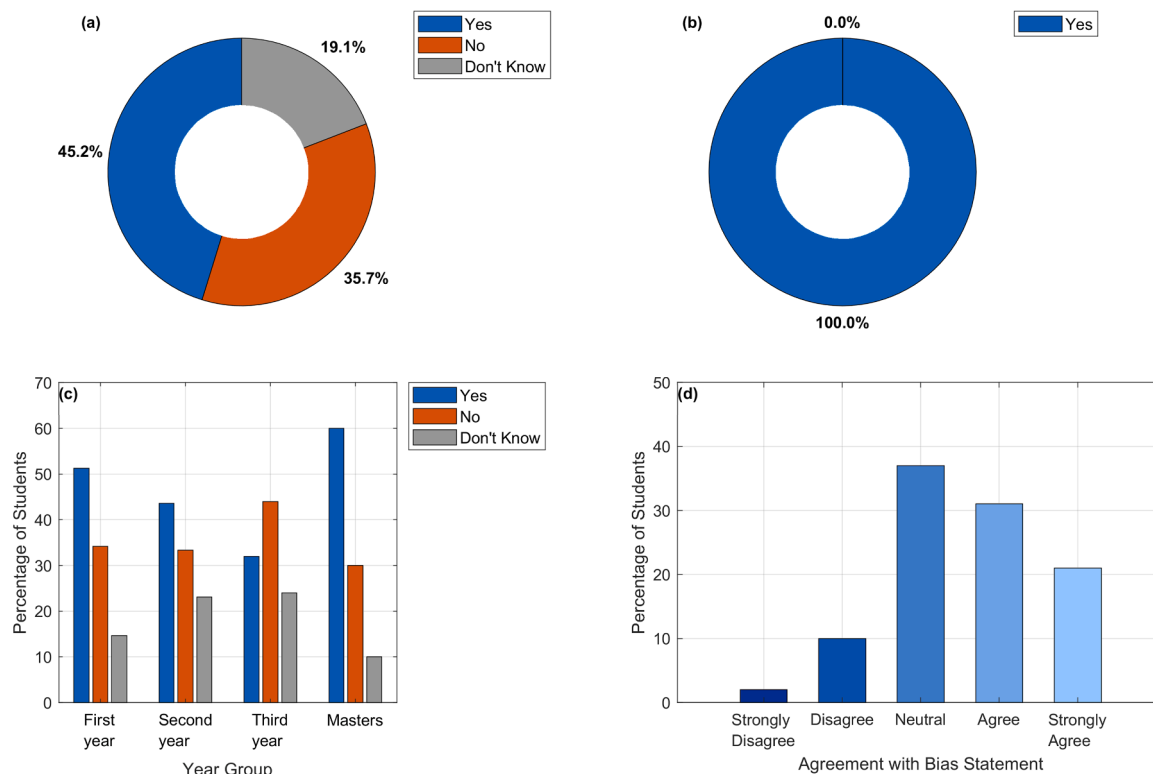


Fig. 1. Responses to the survey question “Do you have any concerns about the use of generative AI in education?” a) student responses (n = 115); b) staff responses (n = 17); c) student split by year group; and d) student responses to the question “To what extent do you agree with the statement ‘Generative AI tools exhibit biases and unfairness in their output’” (n = 115).

diversity. Staff highlighted this issue more prominently, raising a need to interrogate outputs and warning of cultural bias reinforcing a “Western worldview.” Students reiterated concerns and identified risks associated with ignoring different perspectives and attitudes (Chen et al., 2023; Hammond et al., 2024).

Further, copyright and infringement of intellectual property emerged as a major staff concern, with many questioning the legitimacy of knowledge creation “without acknowledgment of royalty,” an issue which was not directly surveyed in the distributed questionnaires but continually surfaced in interviews. This reinforces the complexity of ethical concerns that aren’t always articulated unless prompted, thus supporting the value of a mixed method approach (Dotan et al., 2024; Zawacki-Richter et al., 2019). Similarly, an ethical issue which was only raised by staff was the potential for equity and accessibility disparities between students from different socio-economic backgrounds. For example, Staff 1 noted that it was “massively unfair that a rich kid can get a better AI model than a poor kid”. This difference in accessibility may drive inequalities in education with those from a wealthier background benefiting from access to improved tools, thus potentially enhancing their work quality, attainment, and hence employability. Accessibility issues have also been previously raised by academic research (Capraro et al., 2024; Giannakos et al., 2025), charities (Say, 2024), and thinktanks (Taylor, 2024).

Finally, environmental concerns also arose unprompted, with staff criticising GAI’s “unholy amounts of energy and water use” and students detailing the need for these issues to be addressed in their education. Given the growing need to emphasise sustainable process design, these issues must be considered in the design of chemical engineering curricula and integration of GAI education into existing courses (Baidoo-Anu and Owusu Ansah, 2023; Batista et al., 2024). Additional quotes evidencing this range of ethical concerns is provided in Table S1 in the Supplementary Materials.

3.2. Theme 2: reliability and accuracy of AI outputs

A threshold for adopting GAI in chemical engineering education lies with the tool’s reliability and accuracy. These attributes shape whether the users view tasks such as summarising literature, generating feedback, or even drafting coursework, as credible uses for AI. In previous research, the need for academic rigour formed staff approaches to GAI, while students tended to focus on more practical applications (Michel-Villarreal et al., 2023; Yilmaz, Yilmaz, et al., 2023). Quantitative data from the student survey is provided in Fig. 2, which indicates that largely, students remain quite sceptical with regards to the reliability and accuracy of AI outputs. In answer to the question “How would you rate the quality of the AI-generated information you have encountered in your studies?”, 53 % of respondents rated it as “Mediocre”, with only one person (<1 %) considering it “Very Good”. Similarly, 59 % of students referred to challenges when using GAI (Fig. 2b), which may cause this foundational distrust. Year-by-year comparisons (Fig. 2c) revealed increased indecisive opinions with academic progression, with 35 % of first-year students rating the outputs positively. This may reflect a more novel interaction with outputs compared to the critical reasoning demanded in later year groups (Farazouli et al., 2024).

From the surveys, student focus groups and staff interviews, participants consistently expressed concerns about GAI’s transparency, reliability, and credibility. Examples of quotes are provided in the Supplementary Materials, Table S2. Both groups explained issues with verifying sources or data, noting risks of “hallucinating citations.” This leads to the technical reliability being undermined as well as causing conflicts with academic and professional standards, which are core requirements of accredited chemical engineering degree programmes, and indeed engineering programmes more widely (Quality Assurance Authority, 2023). Further, staff and students also highlighted that AI often lacked the cognitive depth for higher-order tasks such as design, integration, and complex reasoning. For example, one student noted that

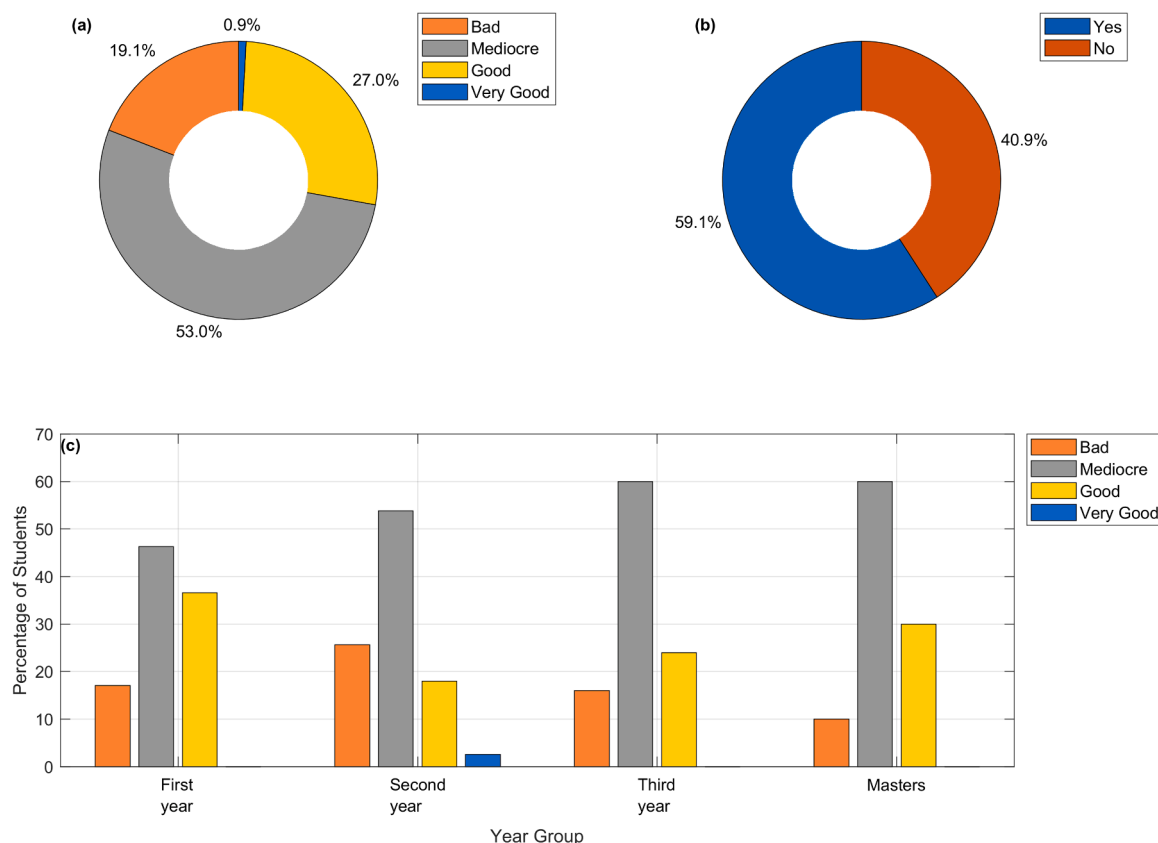


Fig. 2. Student responses ($n = 115$) to a) "How would you rate the quality of the AI-generated information you have encountered in your studies?"; b) "Have you faced any challenges or limitations when using generative AI tools in your studies?"; and c) breakdown of responses to quality of AI output by year group.

"For more complex problems, AI tends to fall over, and either produce rubbish or go around in circles", while staff say that "in aspects of design justification, it still struggles". In addition, staff members in particular also explained that even though LLM outputs often appeared confident and complete, "they break down under close examination". This apparent confidence creates gaps in fundamental reasoning, potentially de-skilling students and reinforcing the need to validate outputs (Farazouli et al., 2024; Tan and Maravilla, 2024). This ability to confidently string together information does not equal genuine understanding, an issue which students could carry forward throughout their education. These findings suggest that reliability issues in GAI are epistemological (i.e. embedded in the nature, origin, and limits of the information presented), and not just purely technical, which hence challenges existing academic standards (Michel-Villarreal et al., 2023).

3.3. Theme 3: impact of GAI on student learning

This theme explores how GAI has impacted student learning, skill development, and confidence. GAI has been perceived to offer opportunities for personalisation, accessibility, and support in education (Michel-Villarreal et al., 2023; Nazar et al., 2024). However, concerns have also been raised over the potential for de-skilling students and a need for competent AI literacy of both staff and students, in order for them to succeed in future work (Farazouli et al., 2024; Tan and Maravilla, 2024).

From a student perspective, the survey revealed strong support for GAI's educational value, with over 65 % reporting GAI as a positive influence on self-learning as illustrated by Fig. 3a, particularly in third- and fourth-year cohorts (Fig. 3b). Similarly, perceived educational development showed mostly positive or neutral responses (Figs. 3c and 3d), a sentiment that was expressed proportionally across year groups. This is somewhat at odds to the responses to previous questions which

examined the perceived quality of AI outputs; here the most popular response was that the quality is "Mediocre", yet despite this, there is still a perceived educational value to AI tools.

Students were also asked what the single most important skill is which they need to develop to use AI tools effectively. The vast majority (60 %) quoted critical thinking as the most important, which far exceeded other options such as data literacy or technical proficiency as shown by Fig. 4a. This suggests that while students welcomed GAI's application, they recognised the need to improve analytical skills to avoid over-reliance (Farazouli et al., 2024; Yilmaz, Yilmaz, et al., 2023). Students reported a high likelihood of continuing to use GAI during their studies (Fig. 4b), and there was also a high proportion of students who felt comfortable using GAI for chemical engineering tasks (Fig. 4c). This reinforces the importance of embedding critical thinking into future curriculum design (Peláez-Sánchez et al., 2024).

To further assess students' perceptions of how generative AI might impact their learning, the survey asked them to indicate to what extent they agreed with the following statements.

1. Generative AI can facilitate collaborative learning
2. Generative AI provides a judgement-free learning environment
3. Generative AI can influence learning in chemical engineering
4. Generative AI can improve self-learning
5. Generative AI can streamline academia
6. Generative AI has the ability to enhance your education

A 5-point Likert scale from Strongly Disagree to Strongly Agree was used. In the subsequent analysis of this quantitative data, a score of 1–5 was applied from Strongly Disagree to Strongly Agree respectively, which enabled the generation of a composite Likert scale (Koo and Yang, 2025), provided in Fig. 5. Taking the mean of these enabled the calculation of the GAI Educational Impact Composite score. To assess internal

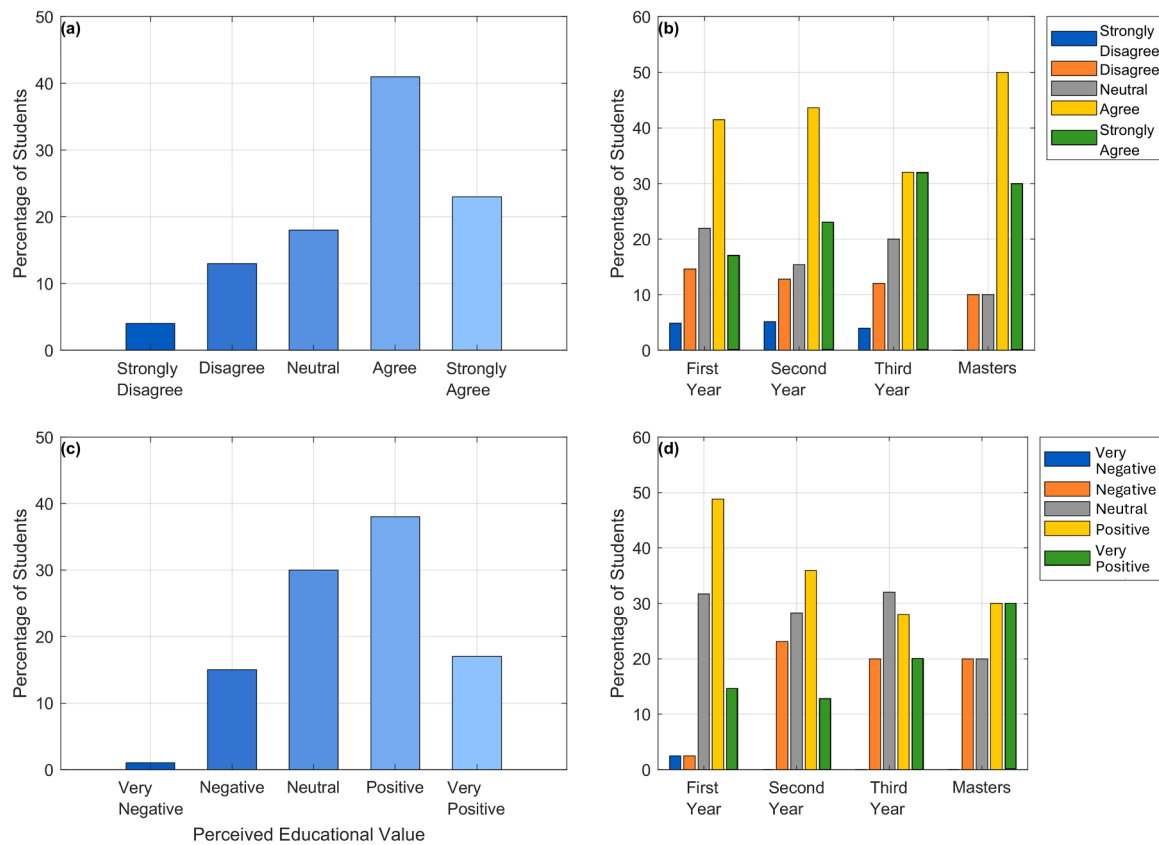


Fig. 3. Student responses ($n = 115$) to a) “To what extent do you agree with the statement ‘Generative AI can improve digital competence and self-learning abilities’”; b) year group split; c) “How would you rate the impact of generative AI in enhancing your own educational development?”; and d) year group split.

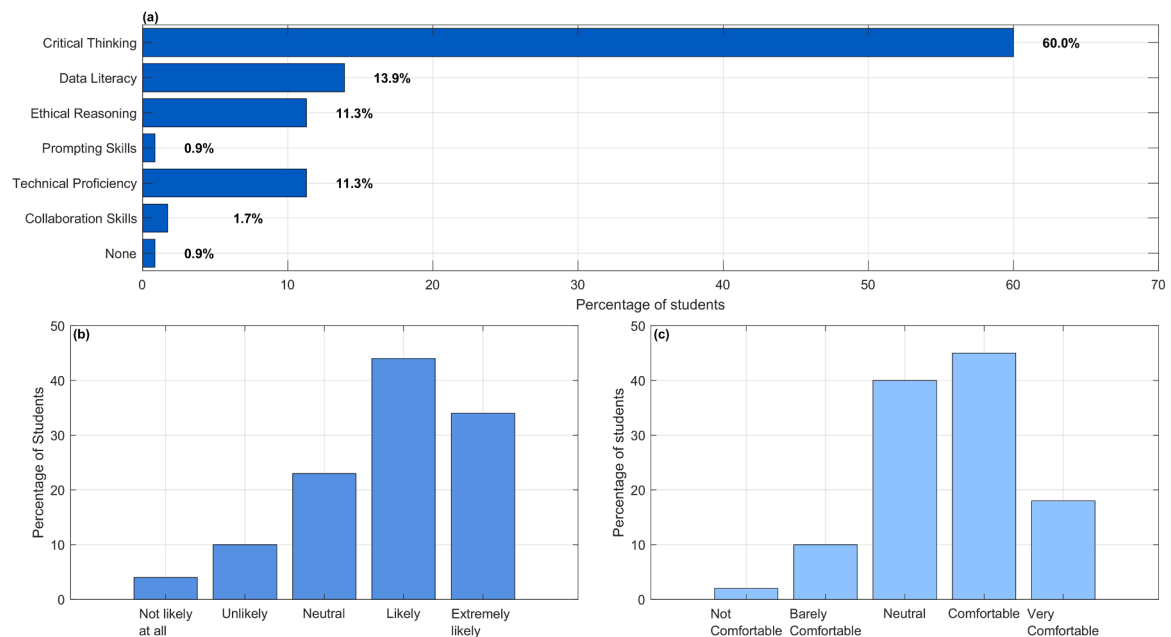


Fig. 4. Student responses ($n = 115$) to the question a) “What is the most important skill you need to develop to effectively use generative AI in your studies?”; b) “How likely are you to continue using generative AI in your studies and future career?”; and c) “How comfortable do you feel using generative AI tools for academic purposes?”.

consistency (i.e. how well these individual questions are measuring the same underlying concept) Cronbach’s α was calculated using Minitab statistical software. This resulted in a value of 0.84 indicating high reliability and thus suggests that students are consistently perceiving

that GAI provides a range of educational benefits from collaborative activities to self-learning, to streamlining workloads.

If students perceive that there are educational benefits to using GAI, then it follows that they are likely to continue using this tool within their

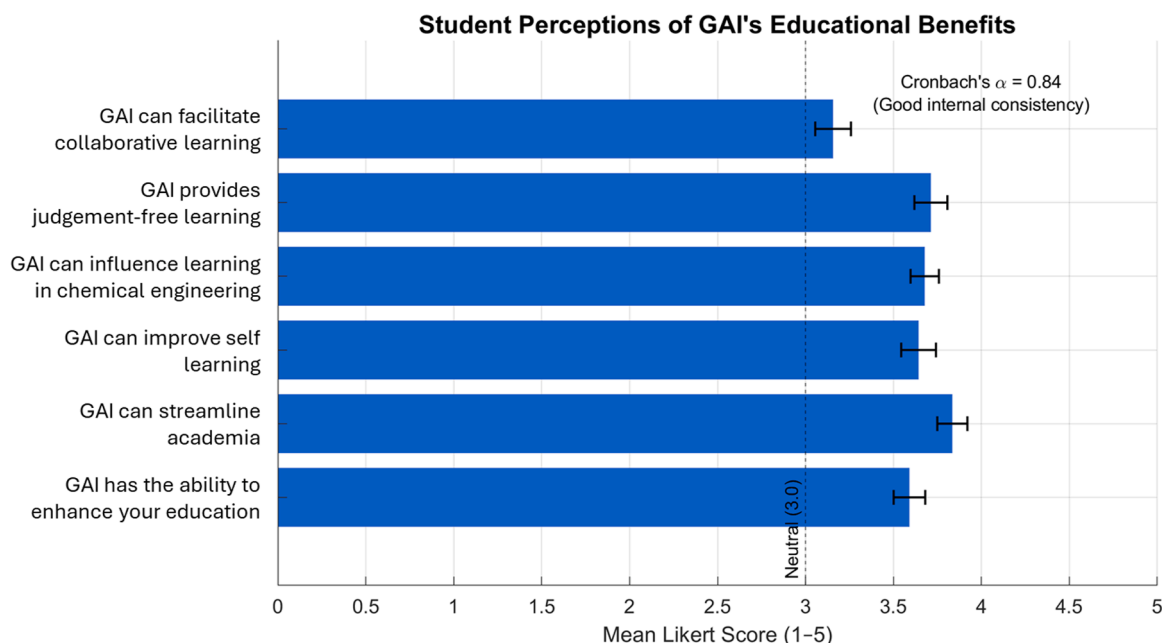


Fig. 5. Composite Likert Scale (Strongly Disagree = 1, Strongly Agree = 5) for all students' survey responses (n = 115).

studies, and beyond. To test this hypothesis, responses to the question “How likely are you to continue using generative AI in your studies and future career?” (data provided in Fig. 4b) were correlated with the composite Likert scale from Fig. 5, and a linear regression model was applied. Fig. 6a shows that the residuals (the difference between the actual observed values and the values predicted by the regression model) do follow a linear relationship ($R^2 = 0.89$). Here, the residuals closely followed the reference line with minor deviations at the tails, which thus demonstrated the suitability of linear regression. A residuals versus fitted values plot (Fig. 6b) showed minor heteroscedasticity, from a slight funnel pattern, but this is unlikely to compromise the robustness of the homoscedasticity assumption. The GAI Educational Impact Composite score was a significant predictor of whether students would continue using this technology ($\beta = 0.80$, $p < 0.001$). Here, the regression coefficient (β) of 0.80 means that for every unit increase in the GAI Educational Impact Composite score (for example, from 3 (neutral) to 4 (agree), the likelihood of students continuing to use GAI will increase by a corresponding amount of 0.80 (for example from 3 (neutral) to 3.80

(very close to 4, “Likely”). The p-value is a statistical significance value which quantifies the likelihood of whether the observed effect is a real result or due to random chance. A score of $p < 0.001$ indicates that there is less than a 0.1 % chance of this correlation happening randomly. These two statistical results explain the finding from the Minitab analysis that the GAI Educational Impact Composite score accounts for 30.5 % of the variation in student's likelihood to continue using GAI. In other words, in this group of 115 students, 30.5 % of the differences in whether they will continue using GAI or not could be explained by how they perceive its educational impact. Although this is a moderate amount, there are clearly other factors which will also impact students' future use which are not captured in this regression model.

From this quantitative data, it can be inferred that students generally take an optimistic view on the use of GAI with the majority agreeing that it can be used to enhance learning. This optimism was reflected in the qualitative data obtained from the survey and focus groups with some noting that GAI offers an “improvement in the quality and accuracy of answers”. Additional quotes are provided in the [Supplementary](#)

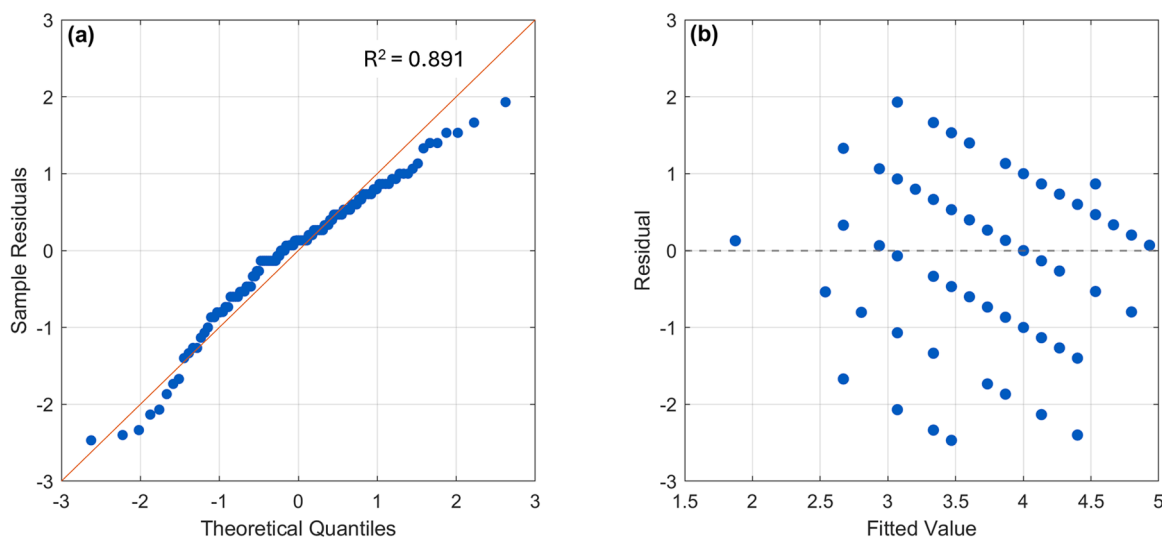


Fig. 6. a) Normal probability plot of regression residuals; and (b) Residuals vs. fitted values plot.

Materials, Table S3. However, staff frequently raised concerns that early-year students may become reliant on GAI before developing foundational skills. This early dependency risks hindering long-term intellectual growth through a scaffolded learning process (Peláez-Sánchez et al., 2024) which is frequently adopted in chemical engineering degree programmes. Despite this, staff did voice that GAI could sharpen critical reasoning skills if “used at both ends” where an AI tool does not act as a tutor, but as a platform from which to improve existing skillsets. Students themselves valued the critical evaluation skills which are garnered through engaging with imperfect outputs, and also repeatedly referenced their ability to upskill themselves in technical areas such as “coding with Matlab”. Finally, students also expressed the need for “working smartly with AI”, and also requested structured AI-literacy training in order to develop specific prompt-engineering skills. This in turn reinforces the need for embedding GAI education into existing curricula, which in turn is likely to have a significant impact on student learning.

3.4. Theme 4: pedagogical and assessment challenges

The widespread adoption of GAI has also raised questions over whether it is necessary to redesign current pedagogical and assessment practices within higher education (Greiner et al., 2023; McDonald et al., 2024). The challenges arising within this theme include plagiarism concerns, which may hence drive an increase in traditional exams, and therefore threaten the continuation of authentic, project-based work. This is a particular concern as this latter assessment method is key to developing a wide range of transferrable skills essential for a chemical engineering graduate’s toolbox (Ravi, 2023; Ravi and Besharat, 2025). Similarly, the student survey data revealed that there is a belief that many people risk over-relying on the use of GAI. As shown in Fig. 7, 81.7% of respondents selected “Agree” or “Strongly Agree” to the question “To what extent do you agree with the statement ‘Students risk over-relying on generative AI?’”. The year-by-year distribution (Fig. 7b), shows that there is a similar perceived risk by students across all stages of the MEng degree programme.

Qualitative data revealed that there is a danger that GAI may impact learning integrity. Staff expressed concerns over authentic project-based assessments becoming redundant in the face of GAI’s ability to simulate work, for example, one staff member stated, “A lot of authentic assessments doesn’t look much more like writing reports, doing analysis, all the stuff that unfortunately ChatGPT is really good at”. In light of this, some staff have redesigned assessment to “require the critique” of GAI-generated answers, in turn, shifting the emphasis of the assessment from production to evaluation which in turn may develop the advanced critical reasoning skills mentioned in Section 3.3. Despite this, there were concerns over plagiarism expressed by both staff and students. As one staff member explained, “It invites cheating left and right,”

reinforcing how unchecked use could undermine academic honesty. Students noted how using GAI without authentic engagement “defeats the point” of coursework. This shows how misuse can lead to a loss of genuine learning and motivation. Additional quotes are provided in the **Supplementary Materials, Table S4.** Although there are clearly assessment-related challenges due to the widespread availability of GAI, particularly LLMs, there was also some optimism in the qualitative responses to surveys, focus groups, and interviews. It was noted that the potential exists for an AI personal tutor especially if a more specialised GAI tool was adapted to chemical engineering content e.g. “Accurately trained AI to provide students with accurate support”. However, participants also strongly stressed that innovations for these tools must receive proper training, rigorous ethical standards, and safeguards to be successfully implemented.

Both staff and students emphasised the need for clear GAI policies, such as a survey respondent who stressed that GAI “requires clear policies and guidelines,” which, without risks creating, or even growing inconsistencies between different modules. This risks unfairness in academic progression, a decrease in academic integrity, and confusion around successful learning development (Johnston et al., 2024; Sime-lane and Kittur, 2025).

Together, these insights highlight how the future of assessment may diverge across two separate paths. Educators may be driven towards an increase in examination to mitigate against the risk of markers inaccurately differentiating between student work and AI-generated content; a risk which has been previously highlighted (Hassoulas et al., 2023). Alternatively, academic practices may evolve, with GAI integrated critically, transparently, and pedagogically, and hence used as a tool to enhance student learning outcomes.

3.5. Theme 5: staff use and application of GAI

Although the focus of the concerns raised so far is on the use of AI-tools by students, the surveys, focus groups, and interviews also highlighted the potential benefit and drawback of staff using and applying GAI in their own teaching practices. Responses showed that while only 23.6 % of staff opposed the integration, nearly 30 % stayed neutral (Fig. 8a), reflecting the slight resistance linked to concerns over assessment integrity and the loss of human interaction. When asked to rate the potential for GAI to enhance teaching and learning (1 being very low, and 5 being very high) and to what extent staff supported the integration of GAI into chemical engineering curricula (1 being very low, and 1 being very high), mid-career staff (11–20 years) exhibited greatest scepticism (Fig. 8b) which may suggest their familiarity with non-AI pedagogical methods, their current workload pressures, and the risks of delegating judgement to GAI. Interestingly, staff with 20+ years’ experience showed the greatest support for GAI integration and most strongly believed in it’s potential. This may be because these more

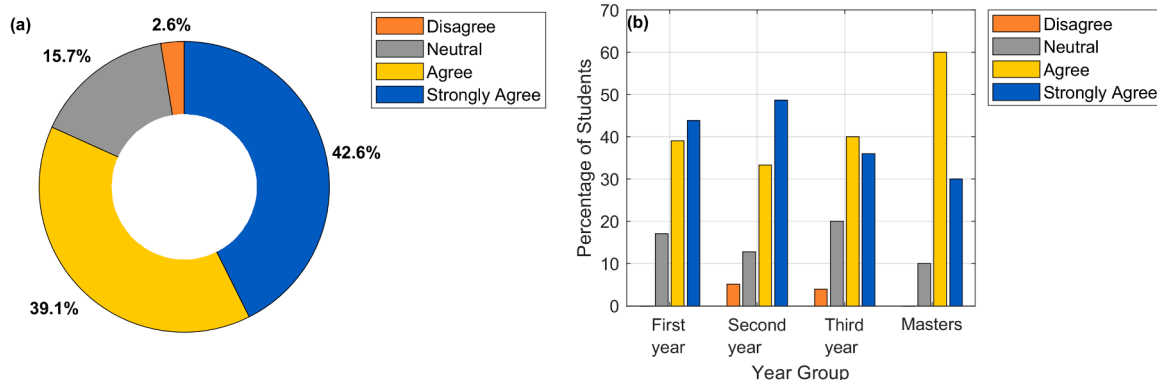


Fig. 7. Student responses to the question “To what extent do you agree with the statement ‘Students risk over-relying on generative AI?’” including split by year group.

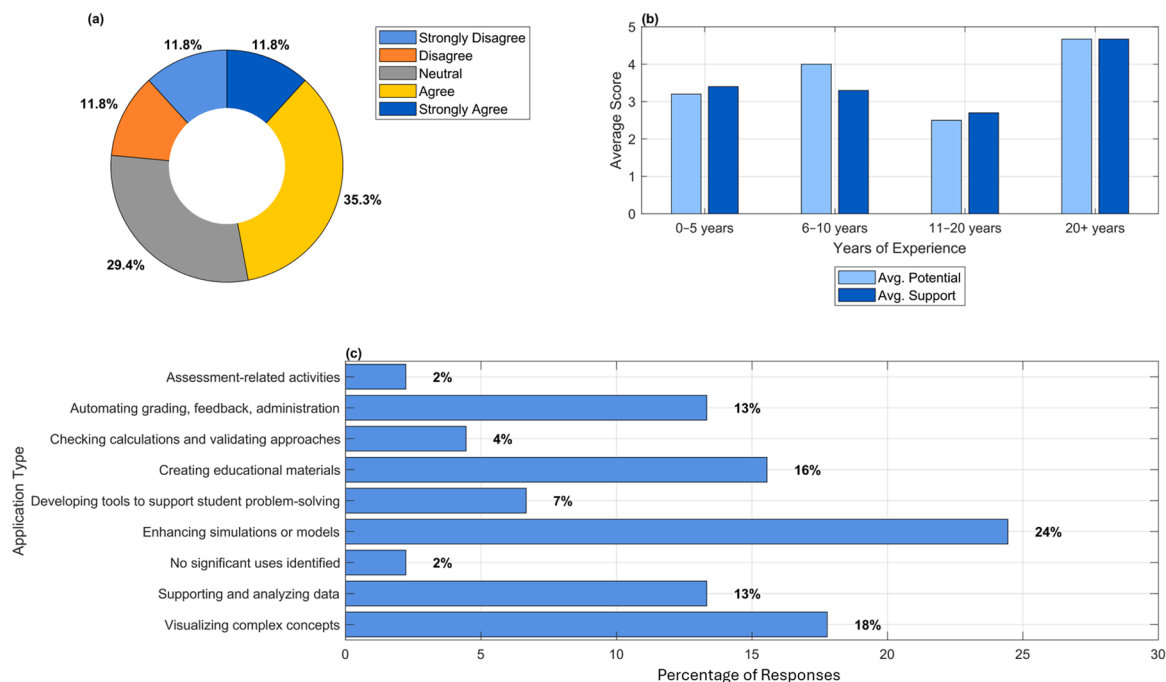


Fig. 8. Staff survey responses (n = 17); a) To the question “To what extent do you agree or disagree with the statement ‘Generative AI tools should be integrated into chemical engineering curricula?’”; b) ratings (1 very low, 5 very high) to the potential for GAI to enhance teaching and learning and support for the integration of GAI tools; and c) main applications of generative AI in chemical engineering education.

experienced staff have worked through the shift from the impact of the

internet, and thus have more support for a new wave of technological

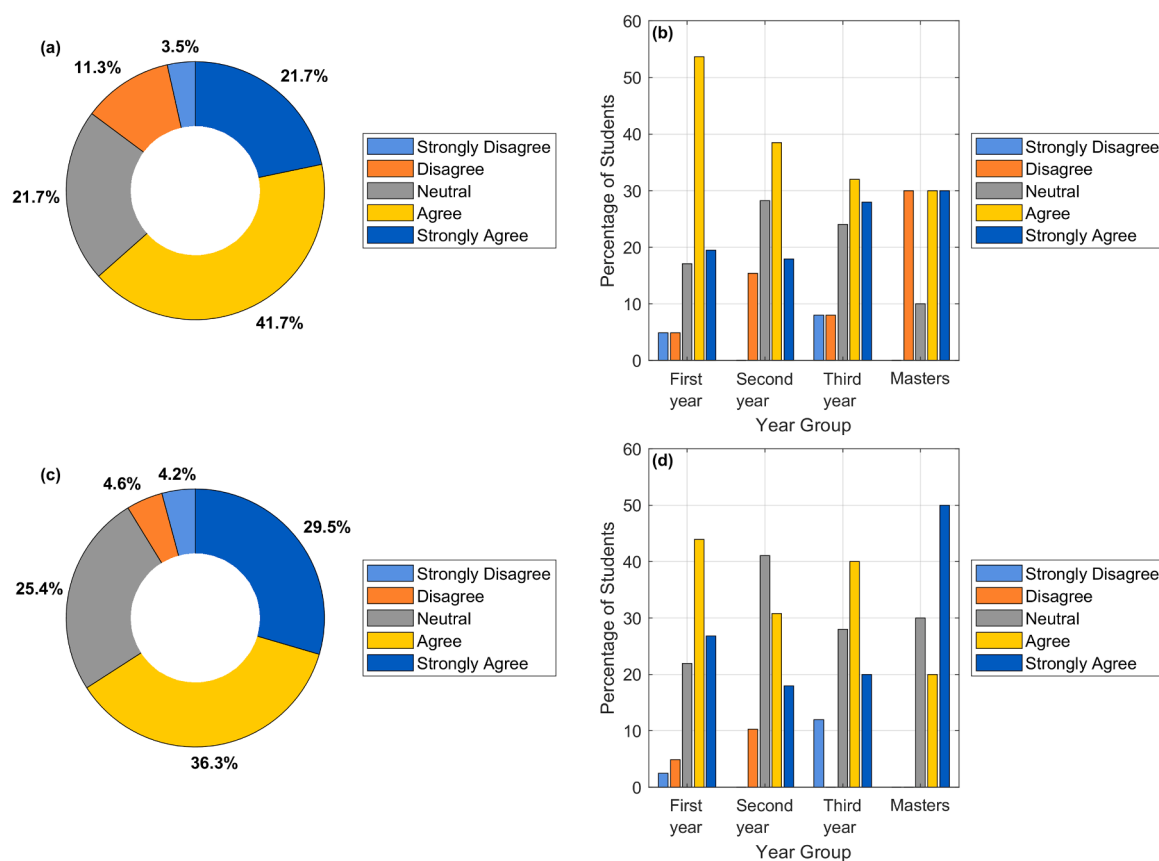


Fig. 9. Student responses to the survey question a) “To what extent do you agree that integrating generative AI will have a positive impact on teaching and learning in the long-term?”; b) split by year group; c) “Do you agree or disagree that generative AI should be integrated into the chemical engineering curriculum?”; and d) split by year group.

development (Batista et al., 2024; McDonald et al., 2024; Peláez-Sánchez et al., 2024). Staff recognised GAI's value for simulations, visualising complex concepts, and teaching material generation (Fig. 8c), with only one person identifying no significant uses. This data shows that, amongst staff, there is generally overall some support for the integration and adoption of GAI into teaching practices. It is worth noting however, that the sample size of 17 represents 31 % of the 54 academic staff who actively deliver some teaching. There may be some selection bias here with those completing the survey more actively supporting or opposing the adoption of GAI tools, while those who are more neutral not participating.

When queried about staff using GAI, students were generally quite supportive, with 63.4 % agreeing that GAI will provide long-term education benefits, as shown in Fig. 9. However, disagreement inclined progressively with academic year groups (Fig. 9b). Amongst students, there was similar support for staff integrating GAI use across the curriculum for all year groups with 65.8 % either agreeing, or strongly agreeing with question “Do you agree or disagree that generative AI should be integrated into the chemical engineering curriculum?”. Together, the quantitative data from staff and students suggest that, although there is not complete support for the integration of GAI technologies, there is overall a preference for its adoption by staff into chemical engineering curricula. This, however, is clearly a complex and nuanced topic, and so the qualitative data, as explored in the following paragraphs, sought to further understand perceptions regarding staff use of GAI and its potential applications.

With regards to staff use, qualitative data from the staff survey and interviews showed that the possibility of GAI to automate administrative tasks and provide feedback to students was generally welcomed. For example, Staff 5 was quoted as “GAI can give all the feedback they [students] need on this is how you should structure your argument, how you should format different aspects, etc. That's fine – I can then focus on the technical”. Additional quotes for this are available in Table S5 in the Supplementary Materials. This suggests that GAI is useful for providing feedback on points which are likely to be deemed trivial such as formatting, but there is still a need for human oversight to make sure that the technical content is described and discussed accurately. This approach as a model for improving feedback has been previously described elsewhere (Chergarova et al., 2023; Nazar et al., 2024). Although a potential benefit, there were also fears that over-automation may create a disconnect between staff and students if relied on too heavily. This could then cause a reduction in the quality of a University's education provision through the displacement of effective academic judgement (Farazouli et al., 2024). Staff raised concerns that reliance may prioritise efficiency instead of understanding, shifting educational values away from cognitive development. Staff did, however, anticipate a slow cultural adoption for integrating GAI into marking and feedback but still objected in principle due to concerns over accuracy, trust, and institutional standards. Similarly, students expressed a dislike of the idea of using AI to grade their assignments with one person stating “I wouldn't want something that involves error to mark my work presently”.

Some benefits to improving workload were acknowledged, but there were concerns that time pressures could negate effective teaching and create knowledge gaps. Whether this risk over-rides the potential benefit of saving time or reducing workload on staff, is likely to be dependent on the individual and their specific workload. Between staff and students, the need for some training to ensure ethical and effective use was consistently emphasised, for example, from the staff survey, one participant responded with “Proper training for educators and students on responsible AI usage will be essential”. A current perceived lack of AI literacy is also likely to act as a barrier to wider integration and adoption (Simelane and Kittur, 2025).

The final finding elicited from the qualitative data which supported the theme of staff use and applications of GAI was that there were differences of opinion regarding AI-generated lecture content and tutorial

materials. Some saw potential (e.g. from the staff survey there was benefit in using LLMs to “generate materials for lecture notes”), whilst others warned “Misused GAI without proper oversight will produce sub-par educational content”. Overall, staff perspectives highlighted that GAI offers the chance for innovation and efficiency, but only if responsible integration includes a human-centred approach to underpin the current quality of chemical engineering education. Therefore, staff support for GAI integration is conditional on the success of its alignment with educational values rather than its technical feasibility.

3.6. Theme 6: the next shift in technology

This final theme was informed by a single code of the same name. Both staff and students were aware that the widespread availability of GAI technologies, and the continued integration into existing software, represents a new shift in technology, akin to the advent of internet and web-based searches. For example, a student in the Year 3 and 4 focus group stated that they “see the release of ChatGPT in a similar light to people who... might have seen the release of Google” whilst Staff 2 stated that there was a “tectonic shift happening”. As technologies develop, the rate of technology development also accelerates, and it may be that we have now reached a tipping point. With the recent release of GPT-5 in August 2025, an even more advanced large language model capable of “PhD-level expertise” that can demonstrate logical reasoning, is now in the public domain (Jamali and McMahon, 2025). This may prove to be a tool with even greater capacity for chemical engineering design, yet from an educational perspective, may present additional challenges as plagiarism becomes harder to detect, the risk of over-reliance increases, and hence the skillsets diminish. Whether this hypothesis is realised is yet to be seen but, as educators, academic staff have a responsibility to ensure that graduates are well-equipped to succeed in an AI-enabled workplace.

The remainder of this section directly compares the perspectives of staff and students, before this research is concluded with a consideration of what impact GAI might have on future chemical engineering education.

3.7. Staff vs. student comparison

The perception of GAI across the School of Chemical Engineering was likely shaped by the role played within the institution, professional expectations, and academic exposure. Although all staff surveyed expressed ethical concerns regarding the use of GAI, students seemed to lack the ability to formally reflect on it to the same detail. They did however intuitively recognise that there were some ethical risks. This difference in appreciation of ethical considerations highlights a development gap, where students seem to operate reactively, rather than systematically (Michel-Villarreal et al., 2023). In turn, this highlights the importance of continued ethical education early within the curriculum.

Although caution in fully trusting AI outputs was expressed by both groups, and a majority of students (59.1 %) thought that there were issues with output quality (Fig. 2b), there was some difference between staff and student perspectives regarding concerns over reliability. Staff were overall much more critical of AI outputs, a criticism which is supported by epistemological standards (Pragya, 2025), whilst students tended to liken GAI to a search engine, particularly early-year students (Farazouli et al., 2024; Greiner et al., 2023). This underscores previously reported concerns regarding AI tools reliability within academic applications (Hammond et al., 2024).

Pedagogically, students framed GAI as a tool for enhancing learning if utilised with guidance. However, this reflects more idealistic use and, when faced with academic pressure and periods of heavier workload, there is a risk that over-reliance on AI-tools increases (Peláez-Sánchez et al., 2024). Staff meanwhile tended to view the integration of AI into chemical engineering curricula through a lens of conditional acceptance: they stressed a need for rigorous control, but as LLMs cannot yet

facilitate higher-order logical reasoning, the ability to critique outputs is essential and hence may improve students' critical analysis skills. To quantitatively compare the difference between staff and students' belief that GAI should be integrated into chemical engineering curricula, an independent samples *t*-test was conducted in Minitab. Both groups were asked the question "To what extent do you agree or disagree that generative AI tools should be integrated in the chemical engineering curriculum?" and given a 5-point Likert scale from Strongly Disagree to Strongly Agree to choose from. A score of 1 was allocated to Strongly Disagree and a score of 5 allocated to Strongly Agree. Due to unequal variances and a large difference in the number of the responses between the two groups, Welch's unpaired *t*-test was used. This provides a more robust alternative to the Student's *t*-test (Delacre et al., 2017). The results of the *t*-test revealed a statistically significant difference in response ($t(19) = -2.2$, $p = 0.04$), with students showing greater support for integrating GAI into curricula than staff. This illustrates that the two means are 2.2 standard errors apart, while the *p*-value of 0.04 represents a 4 % chance that the difference is due to purely random chance and is just less than the typically used 0.05 threshold. This difference is shown graphically in Fig. 10a. Similarly, the calculated Cohen's *d* of 0.68 indicates a moderate to large effect size which confirms the difference in attitude between staff and students.

Qualitative analysis of the interview and focus group data further revealed differences in expectations. Students perceived GAI as a pseudo-personal tutor, with the potential to provide instant feedback and flexible academic support. However, staff focused on safeguarding accreditation standards and maintaining high quality expectations, not thought to be feasible from GAI tools. This reinforces both the attitude and epistemological differences, reflecting different positions within higher education (Kim et al., 2025). Students prioritise the more immediate functionality of new technological tools, while staff emphasise the need for upholding academic rigour and professional standards (Michel-Villarreal et al., 2023). This dynamic reveals a constructivist tension: students seek tools to scaffold their learning (in ideal settings), while staff prioritise tools that reinforce the development of critical reasoning instead of bypassing this growth. The strategic challenge is therefore not to collapse these differences but how to guide them. The support of chemical engineering staff must therefore provide a structure for students to integrate this innovation with integrity and critical literacy (Nazar et al., 2024).

4. Future of GAI in chemical engineering education

4.1. Reframing Pedagogy in an AI-enabled world

Introducing GAI challenges current perceptions and the foundations of teaching and learning. Rather than simplistic technological upgrades, GAI instigates an examination of pedagogical models which raises questions around ethical literacy, active learning, and the role of educators (Luckin et al., 2016). Based on the quantitative and qualitative data presented above, six key considerations have been identified below which are essential for the integration of GAI tools into chemical engineering curricula.

1. Ethical literacy gaps. The lack of articulation from students regarding ethical concerns compared to staff highlights the need for formalised ethical education regarding AI use.
2. Point of integration. GAI education need should be integrated into early-stage curricula to instigate attitude shifts from university being a pass/fail paradigm to one focused on learner development. Staff concerns that early-year students may become over-reliant on GAI, and build up a "debt" reinforces the need for this shift.
3. Reliability challenges. Students view critiquing GAI outputs as a developmental skill, as trust within GAI outputs was rated by 53 % of students as "mediocre". Methods for how staff can nurture this skill must therefore be considered when designing educational activities.
4. Constructivist alignment. GAI can enhance analytical engagement if appropriately utilised, but risks undermining cognitive development if left unstructured thus highlighting the need for intentional curriculum design. Students imagine the opportunity to start "working smartly with AI" and staff comment that it should become a platform to improve learning, supporting constructivist framing.
5. Risk of over-reliance. When scaffolded effectively, GAI supports Vygotsky's Zone of Proximal Development (Vygotsky, 1980) by offering cognitive support without displacing intellectual growth. Students described GAI as a helpful aid but also warned against overuse, validating the potential for GAI to become a crutch (Section 4.3.3), which would undermine development, creating learning deficits.
6. Need for critique. The constructivist lens must extend to assessment practice and create tasks which require critical engagement to ensure that the evaluation of outputs reinforces development (Boud and Molloy, 2013; Moorhouse et al., 2023). This work has shown that some staff are starting to redesign assessments which explicitly

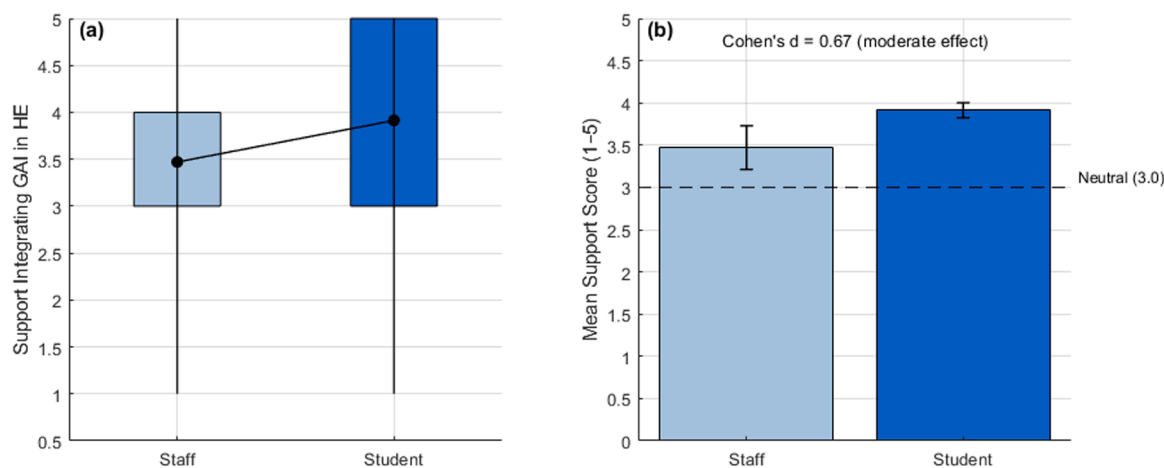


Fig. 10. Comparison of staff and student responses to the question "To what extent do you agree or disagree that generative AI tools should be integrated in the chemical engineering curriculum?" (1 = Strongly Disagree, 5 = Strongly Agree). a) boxplot distribution; and b) comparison of means with error bars showing standard error.

include GAI critiques to mitigate against some integrity issues, whilst providing GAI education.

From a TPACK perspective, students' operational proficiency with GAI is not sufficient for meaningful integration. As detailed in 3.5, staff supported GAI use only where it aligned with values, raising concerns over a threat to content quality. Staff must therefore guide students to foster these critical, ethical, and contextual understandings to embed GAI within chemical engineering teaching formats (Al-Abdullatif, 2024). There is clearly a need for human-centred education; GAI's disruption is philosophical as well as technical, challenging traditional approaches to knowledge construction. Student responses in Section 3.3 reinforced the need for clear guidance and AI literacy support, compared to staff in Section 3.5 who resisted GAI due to a perceived loss of relational education. Dialogue, mentorship, and intuitive feedback (relational pedagogies) must remain core priorities (Tan and Maravilla, 2024). Within higher education in particular, the role of academic staff is not just to transfer knowledge but a facilitator to support conceptual development. The unique value of human-centred education, epistemological depth, and professional judgement must be continually emphasised to mitigate potential tensions. Educators must reposition themselves as mentors in critical learning experiences, to actively shape how students learn with GAI compared to students viewing staff as just sources of knowledge.

4.2. Strategic implications for GAI integration

Progressing with GAI integration requires a number of actionable steps. From this research, three core activities have been identified:

1. Embedding AI Tools and Literacy: Critical thinking, ethical reasoning, and technical competency must become foundational pillars within chemical engineering education.
2. Actively Support Constructivist Use: Integrating GAI into learning objectives, assessment design, and academic support structures will ensure the maximisation of a tool providing efficiency improvements to current practices.
3. Align Academic Practice with Professional Contexts: Educational programmes must evolve towards applied and reflective AI competencies which align with industry expectations.

Trust in GAI outputs is pivotal. Students embraced GAI for formative learning, but rejected its initial introduction for summative assessments, including their marking. This shifts the debate on integration to the accountability and authority of GAI compared to staff. Students' rejection of GAI here reflects a deeper reluctance to delegate authority to non-human systems, revealing distinctions between tools that support learning and those that risk undermining intuitive, responsive feedback and the trust central to human-centric teaching. Institutions must therefore embed AI within critical, context-sensitive frameworks to ensure that students develop skills in bias / error recognition and correction. Rather than retreating to exam-based assessment (a pragmatic but potentially regressive response) innovation must occur. Trialling AI-resilient strategies, such as critiquing GAI outputs, may initially succeed, but workarounds will emerge, especially with the recent release of GPT-5 (Jamali and McMahon, 2025). Assignments which evoke layered validation offer pathways that preserve educational value while prompting critical engagement and deep learning. The danger of dependency is a consistent issue. Staff warned that over-reliance on GAI could minimise effective critical thinking and relational pedagogies, threatening long-term educational development. The danger of cognitive debt reiterates the need to effectively integrate critical and reflective learning processes from the start of the curriculum. However, despite this risk, students and staff both welcomed GAI as a personalised tutor, which could offer revision support if grounded in ethical design and trained on specific content around exact pedagogical

framing. To achieve this, GAI must be treated only as a supplement to human-centred learning.

Institutionally, sustainable GAI integration demands that:

1. GAI is reframed as a supplementary aid which enhances staff expertise rather than replacing their professional judgement regarding teaching and learning
2. Differentiated teaching is supported, possibly by applying frameworks like TPACK to effectively connect a new wave of technology to current curriculum approaches.
3. Relational pedagogies are protected. Dialogue, mentorship, and feedback must remain central to relational education.
4. Structured policies are developed to ensure that integration strategies are designed with both staff and student input to build coherent developments in chemical engineering education.

Staff responses highlight that GAI creates tension over innovation against rigour, efficiency versus authenticity, and standardisation opposed to autonomy. The conditional support from staff reflects the need to integrate GAI while upholding educational integrity, human expertise, and ethical responsibility.

5. Conclusion

This research explored the integration of GAI into chemical engineering education using a mixed methods approach, highlighting tensions across ethical, epistemic, and pedagogical themes. Both students and staff detailed different priorities regarding the implementation of GAI, although each expressed a cautious optimism. Students emphasised GAI's potential to enhance learning and improve critical thinking skills, due to the critique of GAI outputs. However, this optimism was tempered due to concerns about over-reliance, a potential loss of foundational skills, and the risk of reducing effective engagement with content.

Ethical concerns were raised by both groups, but staff raised more detailed unprompted points compared to the abstract concerns of students. Staff also called for a stronger institutional framework to uphold academic integrity and ensure responsible usage. Trust remained moderate: students raised doubts over the reliability of content and staff explained how outputs didn't meet academic standards in chemical engineering. Pedagogically, staff viewed project-based assessments at a point of risk which will require the restructuring of assignments, while fearing a quick return to traditional exams.

This research contributes to the growing literature on AI in higher education by focusing on the specific context of chemical engineering, incorporating both student and staff perspectives. It uncovers practical, philosophical, and pedagogical challenges in GAI's integration through the lens of constructivism and TPACK frameworks. By identifying specific year-group perspectives, policy gaps, and pedagogical opportunities, this study offers actionable insights for chemical engineering curriculum designers, which are likely also applicable across a wide range of disciplines.

However, this study has limitations, including the relatively small staff sample size ($n = 17$) and the variations in students' exposure to GAI, especially in the first year who may not have had meaningful exposure yet. Future research should explore the long-term impact of GAI on student skill development and academic performance, particularly in critical thinking and ethical reasoning. Institutional studies should compare GAI integration into engineering curricula under different accreditation standards and governance models to allow benchmarking and identify the best scalable practices. Finally, continued investigation into the ethical and equity impacts of institutional GAI rollouts will be essential to understand the impact on trust and fairness in AI-enhanced education.

As GAI becomes increasingly embedded in higher education, the role of the educator will likely shift from acting as a primary knowledge

transmitter to a more specialised learning facilitator. This transformation will require staff to become competent or trained in how to use GAI as well as requiring staff to model ethical engagement and critical evaluation for students. Institutions must promote assessment models which require interpretation, justification, and critique instead of output production to support cognitive development with GAI instead of substitute it.

The introduction of GAI also highlights a culture shift in how students perceive learning. Traditional models which value repetition and knowledge retention must evolve towards reasoning, adaptability, and digital fluency as GAI is increasingly integrated. Pedagogical strategies must aim to preserve intellectual integrity but start to embrace the flexibility, efficiency, and opportunities which GAI offers to assessment and curricula design. Hence, there is a critical need for thoughtful governance and policy development to enable GAI to become a tool to enhance learning outcomes and skill development, and not a hindrance to effective chemical engineering education.

CRedit authorship contribution statement

Matthew J. Keith: Writing – original draft, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Evan Cunningham:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Phillip Robbins:** Writing – review & editing, Validation. **Iain Kings:** Writing – review & editing, Validation.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ece.2026.100503](https://doi.org/10.1016/j.ece.2026.100503).

Data Availability

Additional survey data is available in the [supplementary materials](#).

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